

Snowflake Assignment Stage, File Format & Transformation

Dataset URL

S3://sfquickstarts/tastybytes/

File/Folder to Process

raw_pos/menu/

Scenario

You are working as a Junior Data Engineer for a food-truck company.

Your task is to load menu data from a public S3 location into Snowflake, clean the data, perform transformations, and create a simple business summary.

Expected flow:

S3 → Stage → File Format → Raw Table → Transformation → Clean Table → Report

Task 1 — Environment Setup

Create:

- One database
- One raw schema
- One clean schema
- One X-Small warehouse

Questions

1. Why is a warehouse required?
 2. Why do we separate raw and clean data?
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Task 2 — Source File Exploration

Use:

S3://sfquickstarts/tastybytes/raw_pos/menu/

Identify the menu file available in this folder.

Find:

- File name
 - File type
 - Delimiter
 - Whether a header exists
 - Number of columns
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Task 3 — File Format

Create a suitable File Format for the source menu file.

Questions

1. What file type did you configure?
 2. What delimiter did you use?
 3. How did you handle the header?
 4. Why is a File Format required?
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Task 4 — External Stage

Create an External Stage pointing to the public S3 dataset.

Questions

1. What is the purpose of a Stage?
 2. Does the Stage actually store the data inside Snowflake?
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Task 5 — Raw Data Loading

Create a raw menu table and load the menu file into it.

The table should include fields such as:

- Menu ID
- Truck Brand Name
- Menu Item Name
- Item Category
- Item Subcategory
- Cost Price
- Sale Price

Record:

- Number of rows loaded
 - Number of files processed
 - Number of errors
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Task 6 — Data Validation

Find:

1. Total number of menu items.
 2. Number of unique truck brands.
 3. Number of unique categories.
 4. Minimum sale price.
 5. Maximum sale price.
 6. Average sale price.
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Task 7 — Transform the Data

Create a clean menu table.

Perform the following transformations:

1. Clean Menu Item Name

Remove unnecessary spaces.

2. Standardize Brand Name

Convert truck brand names to uppercase.

3. Standardize Category

Convert item categories to uppercase.

4. Calculate Profit

5. Calculate Profit Percentage

6. Create Price Category

Apply:

- Below 5 → BUDGET
- 5 to below 10 → STANDARD

- 10 to below 15 → PREMIUM
- 15 and above → LUXURY

7. Create Display Name

Combine:

Truck Brand Name - Menu Item Name

8. Add Load Timestamp

Store the transformation timestamp.

Task 8 — Business Analysis

1. What are the 5 most expensive menu items?
 2. What are the 5 most profitable menu items?
 3. Which truck brand has the highest average selling price?
 4. Which truck brand has the highest average profit?
 5. How many menu items belong to each category?
 6. How many items belong to each Price Category?
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Task 9 — Data Quality

Find the number of records with:

- Missing Menu Item Name
- Missing Category
- Missing Cost Price
- Missing Sale Price

Create a simple quality status:

- All important values available → GOOD
 - One value missing → REVIEW
 - Multiple values missing → POOR
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Task 10 — Final Summary

Create a truck-brand-level summary containing:

- Truck Brand Name

- Total Menu Items
- Average Sale Price
- Average Profit
- Minimum Sale Price
- Maximum Sale Price

Identify the most profitable truck brand.

Submission

Submit:

- Snowflake worksheet
- Source-file identification screenshot
- Successful loading screenshot
- Raw table screenshot
- Clean table screenshot
- Profit calculation result
- Price-category result
- Final brand summary
- Answers to the conceptual questions